

Lecture #30

Content to be covered

- ❖ Thermal radiation
- ❖ Blackbody
 - ❑ Black body Radiation
- ❖ Rayleigh-Jeans Law
 - ❑ Ultraviolet catastrophe
- ❖ Wien's Law of Radiation

Resources to be consulted

- ❑ Concepts of Modern Physics by Arthur Beiser
- ❑ http://phet.colorado.edu/sims/blackbody-spectrum/blackbody-spectrum_en.html

Thermal radiation: Characteristics

- ❑ An object at any temperature emits radiation sometimes referred to as thermal radiation. Characteristic of this radiation depends on the temperature and the properties of the object.
- ❑ At low temperatures, the wavelengths of thermal radiation are in the infrared region, as the temperature increases the objects begins to glow red means enough visible radiation is emitted so that object appears to glow.
- ❑ As temperature increases, the thermal radiation it emits consists of a continuous distribution of wavelengths from the infrared, visible and ultraviolet portions of the spectrum.
- ❑ From classical view point thermal radiation from accelerated charged particles in the atoms near the surface of the object. The thermally agitated particles can have a distribution of accelerations which accounts for the continuous spectrum of radiation. This **classical theory** was inadequate.
- ❑ The basic problem was in understanding the observed distribution of wavelengths in the radiation emitted by a blackbody.
- ❑ As the temperature increases, distinct behavior are observed. Peak of the distribution shifts to shorter wavelengths.

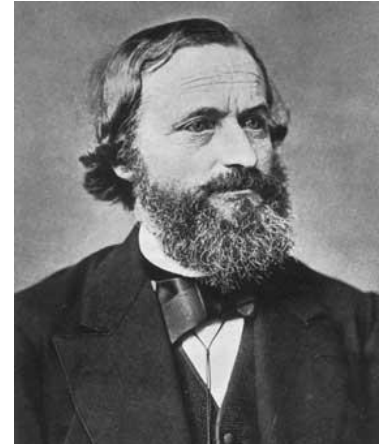
Application of heat radiation

1. White clothes are preferred in summer and dark colored in winter.
2. Cooking utensils are blackened at the bottom and polished on the upper surface.
3. Hot water pipes and radiators used in rooms are painted black so that they can radiate maximum amount of heat to the room. The same pipes outside the rooms are painted white so that they do not lose heat to the surroundings.
4. Polished reflectors are used in electric heaters to reflect maximum heat in the room.

Blackbody (Need not appear to be black)

A material is constantly exchanging heat with its surrounding (to remain at a constant temperature):

- ❑ It absorbs and emits radiations.
- ❑ Problem: it can reflect incoming radiations, which makes a theoretical description more difficult (depends on the environment)



Gustav Kirchhoff

Blackbody (A perfect emitter and a perfect absorber of radiation)

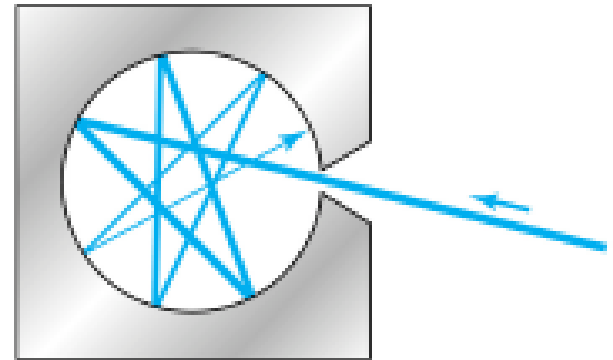
- ❑ Incoming radiations is totally absorbed regardless of the frequency and none is reflected.
- ❑ Sun and Earth's surfaces behave approximately as blackbodies.

The light emitted by a black body is called **black-body radiation**.

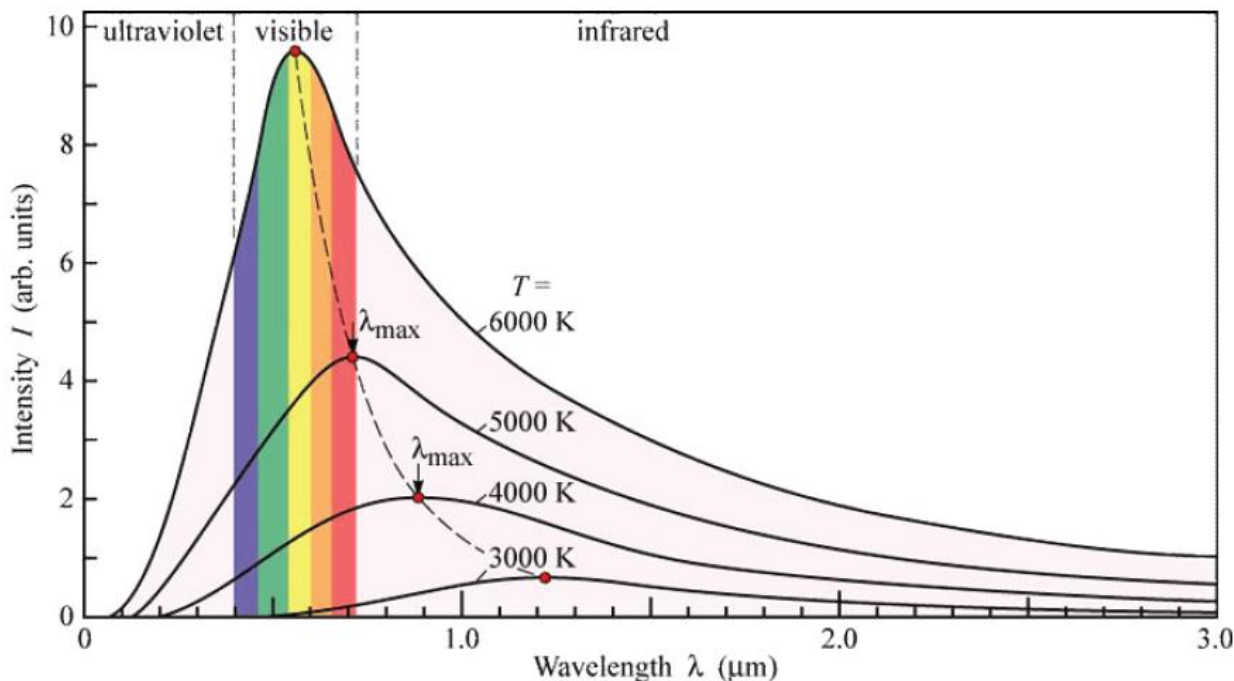
http://phet.colorado.edu/sims/blackbody-spectrum/blackbody-spectrum_en.html

Black body

A black body is an ideal body which allows the whole of the incident radiation to pass into itself (without reflecting the energy) and absorbs within itself this whole incident radiation (without passing on the energy). This property is valid for radiation corresponding to all wavelengths and to all angles of incidence. Therefore, the black body is an ideal absorber of incident radiation.



Black body radiation curves showing peak wavelengths at various temperatures.



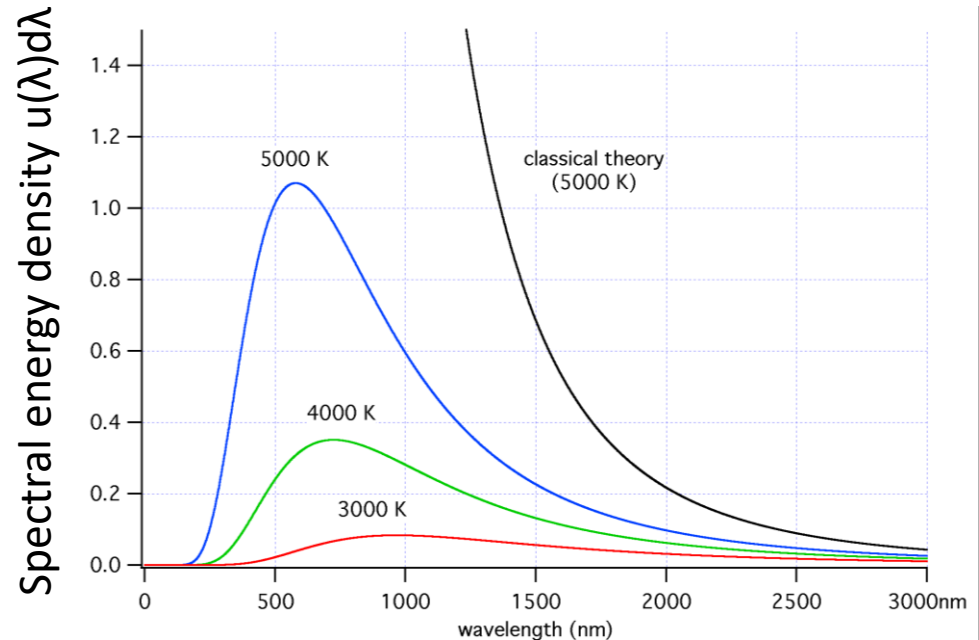
Black body Radiation

- Continuous spectra

Examples-

- Star spectra are nearly Black body
- Radiation out of an oven or a blast furnace
- The Universe
- Anything which is "hot" will absorb and emit electromagnetic radiation.

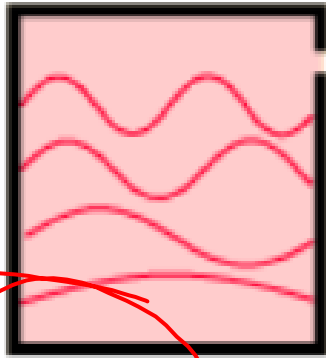
At room temperature, black bodies emit IR light, but as the temperature increases past a few hundred degrees Celsius, black bodies start to emit at visible wavelengths, from red, through orange, yellow, and white before ending up at blue, beyond which the emission includes increasing amounts of UV



Rayleigh-Jeans Law

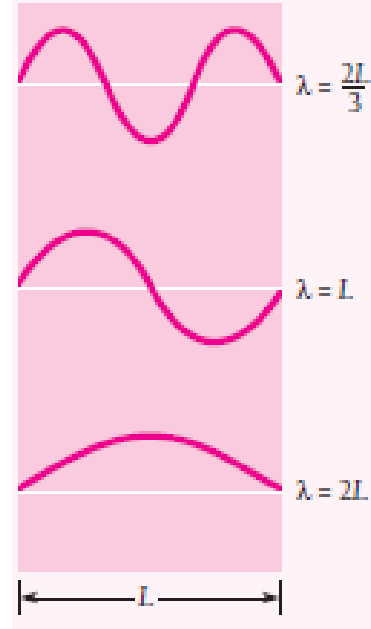
Rayleigh and Jeans considered the radiation inside the black body (cavity) to be a series of standing electromagnetic waves, on the assumption that the *em* wave radiation spectrum emitted by a black body continuously vary in wavelengths from zero to infinity.

The number of standing waves per unit volume (density of em standing waves or allowed modes or density of states) are



$$n(\lambda)d\lambda = \frac{8\pi d\lambda}{\lambda^4} \quad \text{or} \quad n(\nu)d\nu = \frac{8\pi\nu^2 d\nu}{c^3}$$

This equation is independent of shape of the cavity. λ is wavelength, ν is frequency and c is speed of light.



First Attempt to explain Radiation curve behavior. (Classical approach)

Rayleigh-Jeans Law

Therefore, energy of standing waves (total energy per unit volume) with wavelength between λ and $\lambda+d\lambda$ per unit volume would be;

$$u(\lambda)d\lambda = \bar{\varepsilon} \cdot n(\lambda)d\lambda$$

Classical average energy
per standing wave $\bar{\varepsilon} = kT$

$$u(\lambda)d\lambda = \frac{8\pi kT d\lambda}{\lambda^4} \quad \text{or} \quad u(\nu)d\nu = \frac{8\pi \nu^2 kT d\nu}{c^3}$$

This equation is the Rayleigh-Jeans formula for black body radiation.

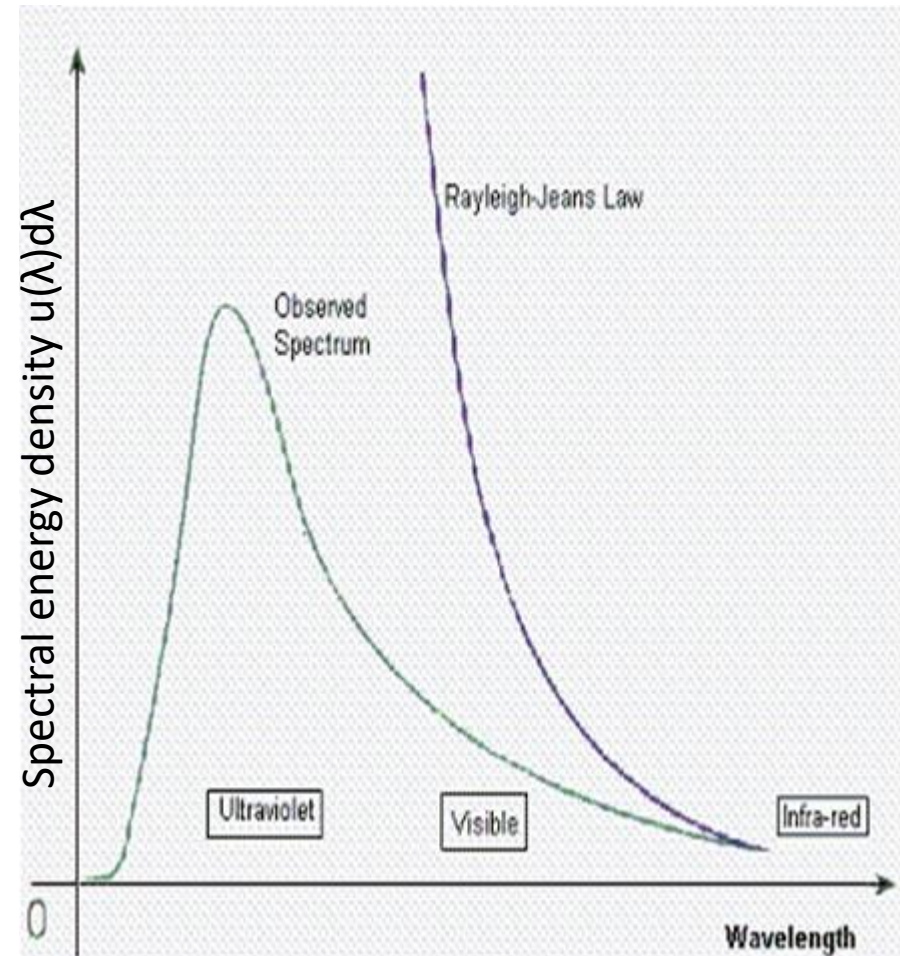
This radiation rate is proportional to this energy density for frequencies between ν and $\nu + d\nu$. This equation contains everything that classical physics can say about the spectrum of blackbody radiation.

How many independent standing waves with wavelengths between 9.5 and 10.5 mm can occur in a cubical cavity 1 m on a side? How many with wavelengths between 99.5 and 100.5 mm?

Rayleigh-Jeans Law

- It agrees with experimental measurements for long wavelengths.
- It predicts an energy output that diverges towards infinity as wavelengths grow smaller.
- The failure has become known as the **ultraviolet catastrophe**.
- Another discrepancy between theory and experiment concerns the total power emitted by the black body.

The **Rayleigh-Jeans law** may be used to accurately describe the long wavelength spectrum of thermal radiation but **fails to describe the short wavelength spectrum of thermal emission**.



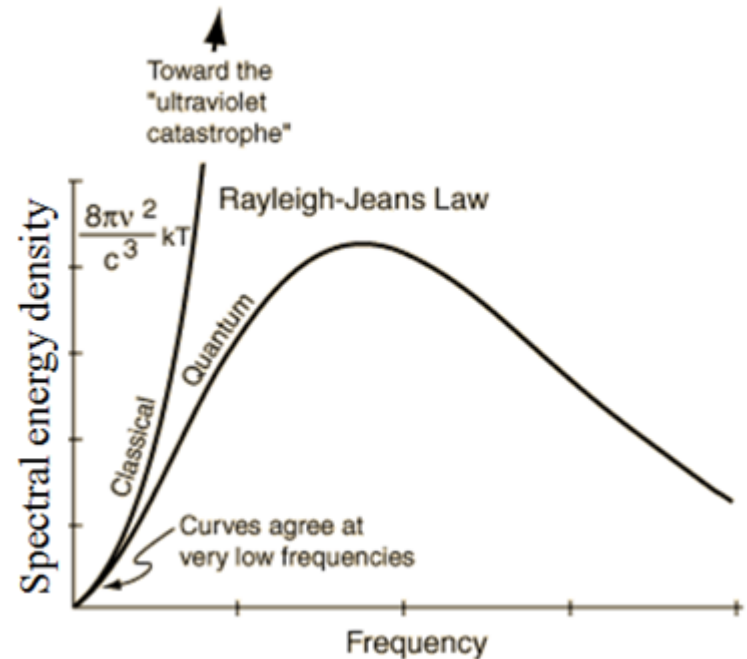
Ultraviolet catastrophe

Let us consider Rayleigh-Jeans formula

$$u(\nu)d\nu = \frac{8\pi\nu^2 kT d\nu}{c^3}$$

According to this equation, as ν increases $u(\nu)d\nu$ increases as ν^2 , and in the limit $\nu \rightarrow \infty$, $u(\nu)d\nu \rightarrow \infty$.

However, in reality as shown in the figure, as $\nu \rightarrow \infty$, $u(\nu)d\nu \rightarrow 0$.

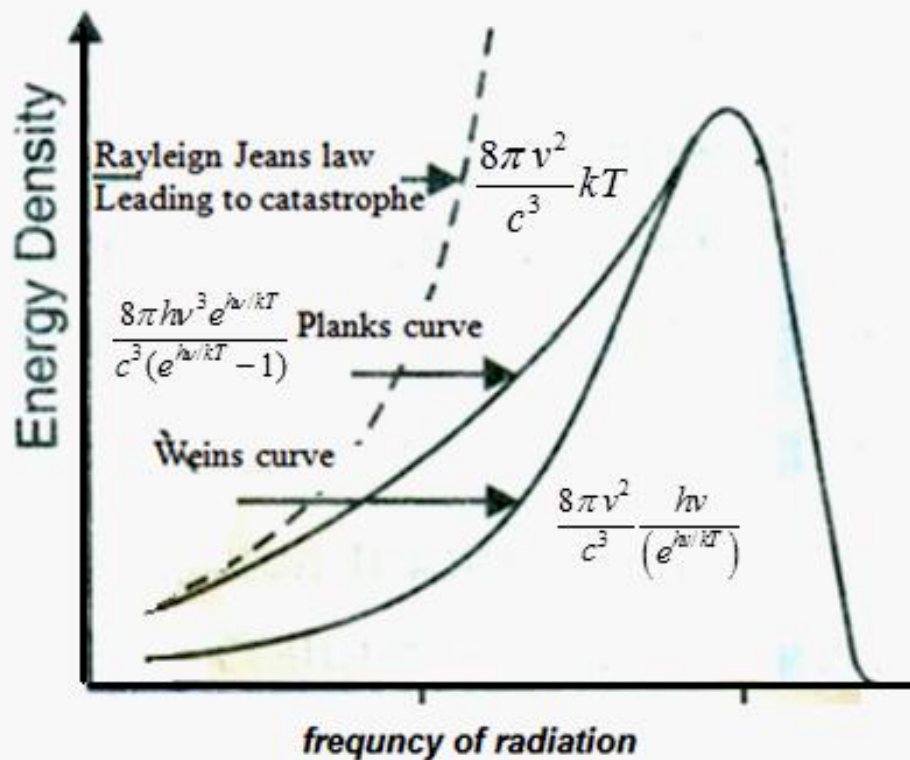


This discrepancy between theory and the experiment is known as “Ultraviolet catastrophe”.

Wein's Law: Second Attempt to explain Radiation curve behavior. (analogy to gas molecules)

Energy = Average energy $\times$ number density

$$E(\nu) d\nu = h\nu e^{-h\nu/kT} \times \frac{8\pi\nu^2}{c^3} d\nu$$



$$E(\nu) d\nu = \frac{8\pi h\nu^3}{c^3} e^{-h\nu/kT} d\nu$$

Called "Wien's Law"